

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY**B. Tech. EE Sem-IV Theory Examination****(April-May,2018)****EE 246 Microprocessor & Microcontrollers****Date: 04/05/2018, Friday****Time: 10:00 a.m. to 01:00 p.m.****Maximum Marks: 70****Instructions:**

1. The question paper comprises two sections.
2. Section I and II must be attempted in separate answer sheets.
3. Make suitable assumptions and draw neat figures wherever required.
4. Comments are necessary in each program.

SECTION-1**Q-1(a)** Draw the pin diagram of 8085 and explain pin function of TRAP and ALE. [5]**Q-1(b)** Explain the following instruction with example (any two). [2]

1. MOV Rd,Rs
2. CMA
3. INR
4. JMP addr

Q-2(a) Explain function of instruction MVI A, 65H, also comment on number of machine cycle, T-state required to execute this instruction and draw timing diagram, instruction is written at memory location 2000H. (op-code for MVI A,data is 3EH) [5]**OR****Q-2(a)** Write an assembly level program for microprocessor 8085 to copy 5 data array starting from D500H to E500H. [5]**Q-2(b)** Write an assembly level program for microprocessor 8085 (any two) [6]

1. To add two 8 bit numbers stored in memory location 5000H and 5001H, and store the answer in register 5002H.
2. To find 2's complement of number stored at memory location A000H and store the answer at memory location A001H.
3. To subtract number stored in memory location 3001H from number stored in memory location 3000h and store answer in register B.

Q-2(c) Fill in the blanks [3]

1. 8085 is ___ bit processor.
2. On hardware reset content of PC will be ____.
3. $(32645)_{10} = (____)_{16}$

Q-3(a) Calculate time required to execute following program. Consider 8085 system frequency of 3MHz. [3]

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MVI B,10H
back: DCR B,
      JNZ back
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OR**Q-3(a)** Write an assembly level program to generate 100micro second delay. Consider 8085 system frequency of 3MHz. [3]

- Q-3(b) Attempt any one** [4]
1. Draw port structure of PORT0 and explain why external pull up resistor is required to send 1 through port pin of PORT0.
 2. Draw port structure of PORT1 and explain why we have to write '1' in port latch to use port pin as an input pin.

- Q-3(C) Attempt any one.** [7]
1. Draw interfacing diagram to interface 16x2 LCD with 8051 and explain pin function of 16x2 LED module. Also, explain step required and write C subroutine to perform write command to LCD.(take necessary assumption and mention each assumptions clearly)
 2. Draw block diagram of 8051 and Explain function of ALU, PC, SP.

SECTION-2

- Q-4(a)** Draw and Explain the Pin diagram for 8051 microcontroller with clear description of function of each pin. [7]

- Q-4(b)** Draw and explain the internal diagram of Timer in mode 1 and mode 2 of operation. [7]

OR

- Q-4(b)** Explain software key debouncing (with program) and Hardware key debouncing (with circuit diagram). [7]

- Q-5(a) Draw and explain function of register listed below. (attempt any three)** [9]

1. PSW
2. TMOD
3. TCON
4. SCON

- Q-5(b)** Write C program to generate 1kHz frequency on pin P1.0 on embedded system build up using P89C51RD2xx and crystal frequency of 12Mhz. [5]

OR

- Q-5(b)** To turn ON LED connected to port pin P2.0 if falling edge appears on $\overline{INT0}$ & Turn it off if falling edge appears on $\overline{INT1}$. [5]

- Q-6(a) Attempt any one.** [2]

1. Write C program to toggle LED connected with port pin P1.7 with large time delay.
2. Write C program to send number 0 to 9 to port 2 continuously with small time delay.

- Q-6(b) Write C program (Attempt any one)** [5]

1. To send "Electrical Engineering" continuously through serial port at baud rate of 9600. Crystal connected to 8051 is working on 11.0592 MHz.
2. To toggle status of pin P1.0 after each 1 second. Crystal connected to 8051 is working on 12MHz.
